

- As we saw in Chapter 2, applications such as e-mail, the Web, and even some network infrastructure services such as the DNS are implemented in hosts (servers) at the network edge. The ability to add a new service simply by attaching a host to the network and defining a new application-layer protocol (such as HTTP) has allowed new Internet applications such as the Web to be deployed in a remarkably short period of time.

4.3 What's Inside a Router?

Now that we've overviewed the network layer's services and functions, let's turn our attention to its **forwarding function**—the actual transfer of packets from a router's incoming links to the appropriate outgoing links at that router. We already took a brief look at a few aspects of forwarding in Section 4.2, namely, addressing and longest prefix matching. We mention here in passing that the terms *forwarding* and *switching* are often used interchangeably by computer-networking researchers and practitioners; we'll use both terms interchangeably in this textbook as well.

A high-level view of a generic router architecture is shown in Figure 4.6. Four router components can be identified:

- *Input ports.* An input port performs several key functions. It performs the physical layer function of terminating an incoming physical link at a router; this is shown in the leftmost box of the input port and the rightmost box of the output port in Figure 4.6. An input port also performs link-layer functions needed to interoperate with the link layer at the other side of the incoming link; this is represented by the middle boxes in the input and output ports. Perhaps most crucially, the lookup function is also performed at the input port; this will occur in the rightmost box of the input port. It is here that the forwarding table is consulted to determine the router output port to which an arriving packet will be forwarded via the switching fabric. Control packets (for example, packets carrying routing protocol information) are forwarded from an input port to the routing processor. Note that the term *port* here—referring to the physical input and output router interfaces—is distinctly different from the software ports associated with network applications and sockets discussed in Chapters 2 and 3.
- *Switching fabric.* The switching fabric connects the router's input ports to its output ports. This switching fabric is completely contained within the router—a network inside of a network router!
- *Output ports.* An output port stores packets received from the switching fabric and transmits these packets on the outgoing link by performing the necessary link-layer and physical-layer functions. When a link is bidirectional (that is,

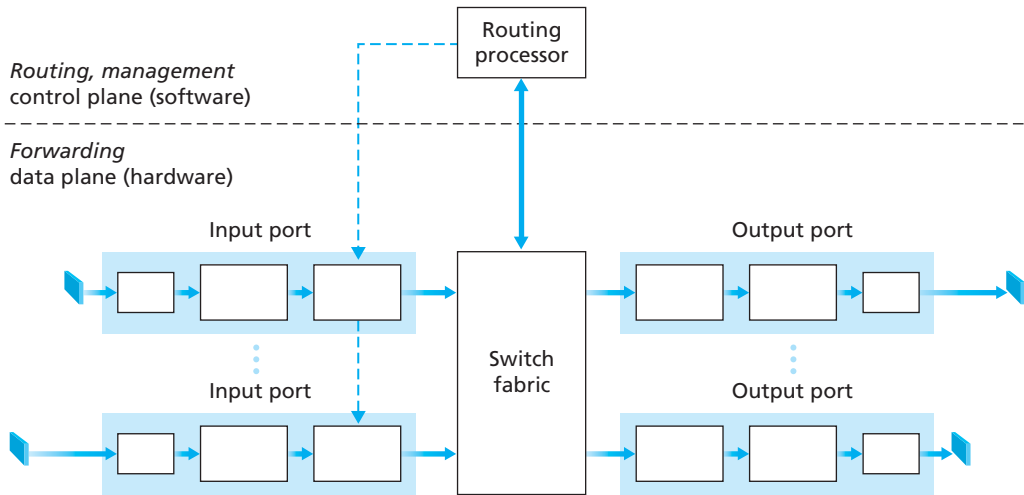


Figure 4.6 ♦ Router architecture

carries traffic in both directions), an output port will typically be paired with the input port for that link on the same line card (a printed circuit board containing one or more input ports, which is connected to the switching fabric).

- *Routing processor.* The routing processor executes the routing protocols (which we'll study in Section 4.6), maintains routing tables and attached link state information, and computes the forwarding table for the router. It also performs the network management functions that we'll study in Chapter 9.

Recall that in Section 4.1.1 we distinguished between a router's forwarding and routing functions. A router's input ports, output ports, and switching fabric together implement the forwarding function and are almost always implemented in hardware, as shown in Figure 4.6. These forwarding functions are sometimes collectively referred to as the **router forwarding plane**. To appreciate why a hardware implementation is needed, consider that with a 10 Gbps input link and a 64-byte IP datagram, the input port has only 51.2 ns to process the datagram before another datagram may arrive. If N ports are combined on a line card (as is often done in practice), the datagram-processing pipeline must operate N times faster—far too fast for software implementation. Forwarding plane hardware can be implemented either using a router vendor's own hardware designs, or constructed using purchased merchant-silicon chips (e.g., as sold by companies such as Intel and Broadcom).

While the forwarding plane operates at the nanosecond time scale, a router's control functions—executing the routing protocols, responding to attached links that

go up or down, and performing management functions such as those we'll study in Chapter 9—operate at the millisecond or second timescale. These **router control plane** functions are usually implemented in software and execute on the routing processor (typically a traditional CPU).

Before delving into the details of a router's control and data plane, let's return to our analogy of Section 4.1.1, where packet forwarding was compared to cars entering and leaving an interchange. Let's suppose that the interchange is a roundabout, and that before a car enters the roundabout, a bit of processing is required—the car stops at an entry station and indicates its final destination (not at the local roundabout, but the ultimate destination of its journey). An attendant at the entry station looks up the final destination, determines the roundabout exit that leads to that final destination, and tells the driver which roundabout exit to take. The car enters the roundabout (which may be filled with other cars entering from other input roads and heading to other roundabout exits) and eventually leaves at the prescribed roundabout exit ramp, where it may encounter other cars leaving the roundabout at that exit.

We can recognize the principal router components in Figure 4.6 in this analogy—the entry road and entry station correspond to the input port (with a lookup function to determine the local outgoing port); the roundabout corresponds to the switch fabric; and the roundabout exit road corresponds to the output port. With this analogy, it's instructive to consider where bottlenecks might occur. What happens if cars arrive blazingly fast (for example, the roundabout is in Germany or Italy!) but the station attendant is slow? How fast must the attendant work to ensure there's no backup on an entry road? Even with a blazingly fast attendant, what happens if cars traverse the roundabout slowly—can backups still occur? And what happens if most of the entering cars all want to leave the roundabout at the same exit ramp—can backups occur at the exit ramp or elsewhere? How should the roundabout operate if we want to assign priorities to different cars, or block certain cars from entering the roundabout in the first place? These are all analogous to critical questions faced by router and switch designers.

In the following subsections, we'll look at router functions in more detail. [Iyer 2008, Chao 2001; Chuang 2005; Turner 1988; McKeown 1997a; Partridge 1998] provide a discussion of specific router architectures. For concreteness, the ensuing discussion assumes a datagram network in which forwarding decisions are based on the packet's destination address (rather than a VC number in a virtual-circuit network). However, the concepts and techniques are quite similar for a virtual-circuit network.

4.3.1 Input Processing

A more detailed view of input processing is given in Figure 4.7. As discussed above, the input port's line termination function and link-layer processing implement the physical and link layers for that individual input link. The lookup performed in the input port is central to the router's operation—it is here that the router uses the forwarding table to look up the output port to which an arriving packet will be